The documentation and process conversion measures necessary to comply with this document shall be completed by 13 February 2014.

INCH POUND

MIL-PRF-19500/510J
13 December 2013
SUPERSEDING
MIL-PRF-19500/510H
14 January 2013

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPE 2N6249, 2N6249T1, 2N6249T3, 2N6250, 2N6250T1, 2N6250T3, 2N6251, 2N6251T1, 2N6251T3,
JAN, JANTX, JANTXV, JANS, JANSM, JANSD, JANSP, JANSL, JANSR, JANSF, JANSG, JANSH, JANHCB,
JANKCB, JANKCBM, JANKCBD, JANKCBP, JANKCBL, JANKCBR, JANKCBF, JANKCBG, AND JANKCBH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to eight radiation levels is provided for JANS and JANKC product assurance levels. RHA level designators “M”, “D”, “P“, “L”, “R”, “F”, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Physical dimensions. See figure 1 (similar to TO-3), figure 2 (JANHC and JANKC), figure 3 (TO-254AA), and figure 4 (TO-257AA).

1.3 Maximum ratings at T_A = +25°C, unless otherwise specified.

Type	P_T T_A = +25°C	P_T (1) T_C = +25°C	$R_{\theta JC}$ (2)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N6249	6.0	175	1.0	300	200	6.0	10	5.0	
2N6249T1	6.0	175	1.0	300	200	6.0	10	5.0	-65
2N6249T3	4.0	(3) 125	1.3	300	200	6.0	10	5.0	
2N6250	6.0	175	1.0	375	275	6.0	10	5.0	to
2N6250T1	6.0	175	1.0	375	275	6.0	10	5.0	
2N6250T3	4.0	(3) 125	1.3	375	275	6.0	10	5.0	+200
2N6251	6.0	175	1.0	450	350	6.0	10	5.0	
2N6251T1	6.0	175	1.0	450	350	6.0	10	5.0	
2N6251T3	4.0	(3) 125	1.3	450	350	6.0	10	5.0	

(1) For temperature-power derating curves, see figures 5 and 6.
(2) For thermal impedance curves, see figures 7, 8, and 9.
(3) For TO-257 devices with typical mounting and small footprint, conservatively rating at 125 W and 1.3°C/W only.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics at T_A = +25°C.

Types	Limits	h_{FE1} I_C = 10 A dc V_{CE} = 3 V dc	C_{obo} V_{CB} = 10 V dc I_C = 0 100 kHz ≤ f ≤ 1 MHz	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	Pulse response	
					t_{on}	t_{off}
			pF		µs	µs
2N6249, 2N6249T1, 2N6249T3	Minimum	10		2.5		
	Maximum	50	500	15.0	2.0	4.5
2N6250, 2N6250T1, 2N6250T3	Minimum	8		2.5		
	Maximum	50	500	15.0	2.0	4.5
2N6251, 2N6251T1, 2N6251T3	Minimum	6		2.5		
	Maximum	50	500	15.0	2.0	4.5

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

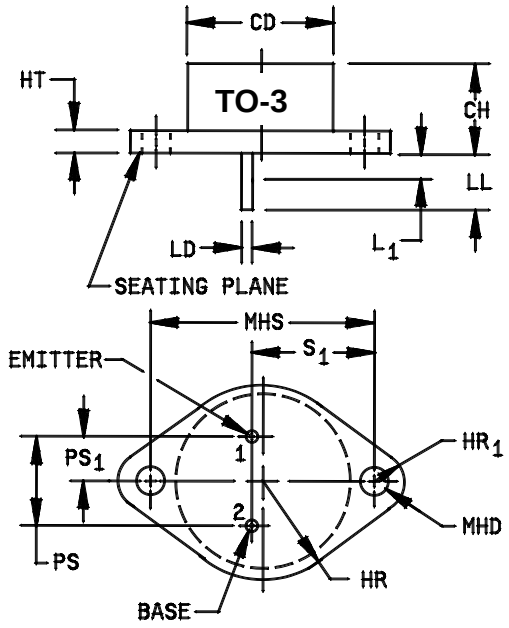
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	
CH	.250	.450	6.35	11.43	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.050	.135	1.27	3.43	
LD	.038	.053	0.97	1.35	3, 5
LL	.312	.500	7.92	12.70	3
L_1		.050		1.27	5
MHD	.151	.161	3.84	4.09	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	2
PS_1	.205	.25	5.21	6.35	2, 3
S_1	.665	.675	16.89	17.15	2

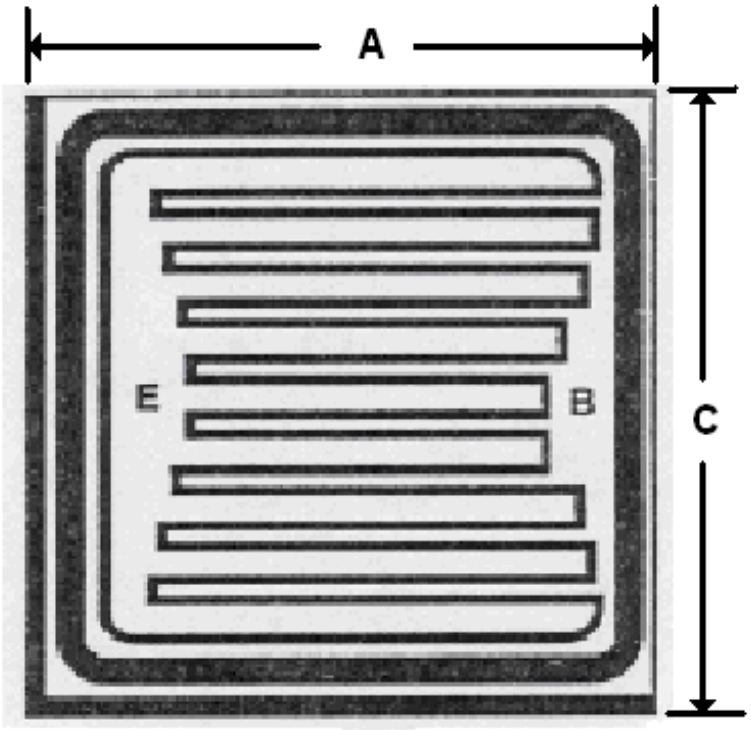


NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. These dimensions should be measured at points .050 (1.27 mm) to .055 (1.40 mm) below seating plane. When gage is not used, measurement will be made at seating plane.
3. Two leads.
4. Collector shall be electrically connected to the case.
5. LD applies between L_1 and LL maximum. Lead diameter shall not exceed twice LD within L_1.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕ symbology.

FIGURE 1. Physical dimensions (similar to TO-3).

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Ltr.	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A, C	.205	.215	5.21	5.46

DESIGN DATA

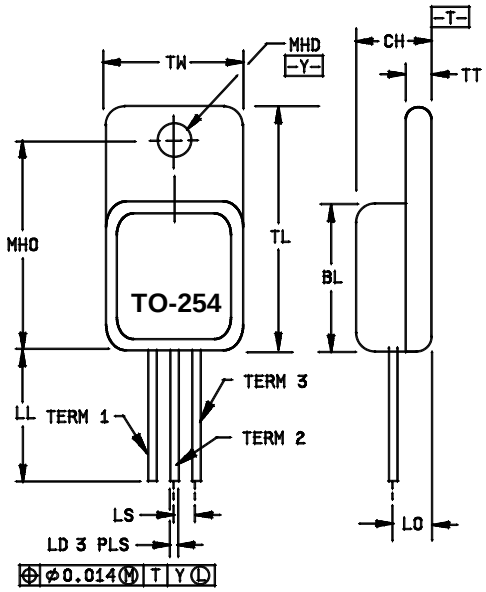
Metallization:

Top:	Aluminum 54,000 Å minimum, 60,000 Å nominal.
Back:	Al/Ti/Ni/Au 10,000 Å minimum, 12,500 Å nominal.
Back side:	Collector.
Chip thickness:	.012 inch (0.305 mm) ±.002 inch (0.051 mm).
Bonding pad:	B = .018 inch (0.46 mm) x .040 (1.02 mm). E = .018 inch (0.46 mm) x .040 (1.02 mm).

FIGURE 2. JANHCB and JANKCB (B version) die dimensions.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.530	.550	13.46	13.97
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.80 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			



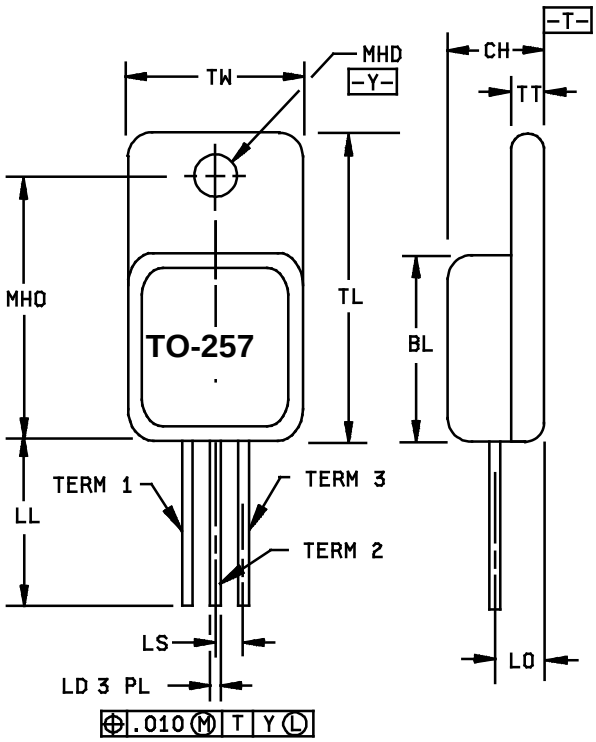
NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Methods used for electrical isolation of terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL203 (ceramic).
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Dimensions and configuration 2N6249T1, 2N6250T1, and 2N6251T1 (TO-254AA).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.410	.430	10.41	10.92
CH	.190	.200	4.83	5.08
LD	.025	.035	0.64	0.89
LL	.500	.750	12.70	19.05
LO	.120 BSC		3.05 BSC	
LS	.100 BSC		2.54 BSC	
MHD	.140	.150	3.56	3.81
MHO	.527	.537	13.39	13.63
TL	.645	.665	16.38	16.89
TT	.035	.045	0.89	1.14
TW	.410	.420	10.41	10.67
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Methods used for electrical isolation of the terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL2O3 (ceramic).
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Dimensions and configuration for 2N6249T3, 2N6250T3, and 2N6251T3 (TO-257AA).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-3), figure 2 (die), figure 3 (TO-254AA), and figure 4 (TO-257AA) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E–IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E–IV of MIL-PRF-19500)	Measurements	
	JANS	JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.1)	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.1)
9	I_{CEX1} and h_{FE1}	
11	I_{CEX1} and h_{FE1} ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater; Δh_{FE1} = ±15 percent of initial value.	I_{CEX1} and h_{FE1}
12	See 4.3.2	See 4.3.2
13	Subgroup 2 and 3 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater; Δh_{FE1} = ±25 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater; Δh_{FE1} = ±25 percent of initial value.
17	For TO-254AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein.	For TO-254AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein.

* (1) Thermal impedance ($Z_{\theta JC}$) limits shall not exceed figures 7, 8, or 9 thermal impedance curves. This test shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in screen 3c and table I, subgroup 2 shall comply with the thermal impedance graphs on figures 7, 8, or 9 (less than or equal to the curve value at the same t_H time) and/or shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10-30 V dc, T_A = +30°C maximum, T_J = +175°C minimum.

4.3.3 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, “Discrete Semiconductor Die/Chip Lot Acceptance”. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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* 4.3.4 Dielectric withstanding voltage.

a. Magnitude of test voltage……..900 V dc.

b. Duration of application of test voltage……..15 seconds (min).

c. Points of application of test voltage……..All leads to case (bunch connection).

d. Method of connection……..Mechanical.

e. Kilovolt-ampere rating of high voltage source…….1,200 V/1.0 mA (min).

f. Maximum leakage current……...1.0 mA.

g. Voltage ramp up time.....500 V/second.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A, all internal leads for each device shall be pulled separately.
* B4	1037	V_{CB} = 10 V dc, adjust power or current to achieve ΔT_J = +100°C.
B5	1027	V_{CB} = 10 V dc; adjust T_A and P_D to achieve T_J = +275°C minimum. P_D = 100 percent of rated P_T minimum; T_A = +100°C maximum.

* 4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
* B3	1037	V_{CB} = 10 V dc, adjust power or current to achieve ΔT_J = +100°C.
	or	
	1027	V_{CB} = 10 V dc; T_J = +175°C minimum; T_A = +30°C maximum.

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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

	Subgroup	Method	Condition
	C2	2036	Test condition A; weight - 10 pounds; time - 15 seconds.
	C3	2016	Non-operating 1,500 G; 0.5 ms, 5 blows in each orientation: X_1, Y_1, Y_2, Z_1 (total 20 blows).
	C5	3131	See 4.5.2. See thermal impedance curves figures 7, 8, and 9.
*	C6	1027 or 1037	V_{CB} = 10 V dc; T_J = +175°C minimum; T_A =+30°C maximum. V_{CB} = 10 V dc, adjust power or current to achieve ΔT_J = +100°C.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANKC types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Group D inspection may also be performed ahead of the screening lot using die selected in accordance with MIL-PRF-19500 and related documents. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 3.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ t_r ≤ +75°C and recorded before the last is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit for $R_{\theta JC}$ see thermal impedance curves figures 7, 8, and 9.

4.5.3 Coil selection for safe operating area (SOA) tests. In selecting coils for use in the clamped and unclamped inductive SOA tests, prime consideration should be given to the recommended commercially available coil. However, due to the extreme critical nature of the coil in these circuits and the wide tolerance of some commercially available coils (+100, -50 percent), it shall be the semiconductor manufacturer's responsibility, to prove upon request, compliance or equivalency of any coil used (commercial or in-plant designed) to within (+20, -10 percent), of the specified inductance at the rated current and dc resistance.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.1	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition B, I_C = 200 mA dc, L = 42 mH, f = 30 - 60 Hz (see figure 10).	$V_{(BR)CEO}$			
2N6249, 2N6249T1 2N6249T3				200		V dc
2N6250, 2N6250T1 2N6250T3				275		V dc
2N6251, 2N6251T1 2N6251T3				350		V dc
Breakdown voltage, collector to emitter	3011	Bias condition B, I_C = 200 mA dc, L = 14 mH, f = 30 - 60 Hz R_{BE} = 50 ohms (see figure 10).	$V_{(BR)CER}$			
2N6249, 2N6249T1 2N6249T3				225		V dc
2N6250, 2N6250T1 2N6250T3				300		V dc
2N6251, 2N6251T1 2N6251T3				375		V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc.	I_{EBO}		100	µA dc
Collector to emitter cutoff current	3041	Bias condition D.	I_{CEO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 150 V dc			1.0	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 225 V dc			1.0	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 300 V dc			1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc.	I_{CEX1}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 225 V dc			10	µA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 300 V dc			10	µA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 375 V dc			10	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 300 V dc			0.5	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 375 V dc			0.5	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 450 V dc			0.5	mA dc
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1).	h_{FE1}			
2N6249, 2N6249T1 2N6249T3				10	50	
2N6250, 2N6250T1 2N6250T3				8	50	
2N6251, 2N6251T1 2N6251T3				6	50	
Collector to emitter saturated voltage	3071	I_C = 10 A dc; pulsed (see 4.5.1).	$V_{CE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			1.5	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			1.5	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			1.5	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; pulsed (see 4.5.1).	$V_{BE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			2.25	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			2.25	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			2.25	V dc
Subgroup 3						
High temperature operation		T_A = +125°C.				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc.	I_{CEX2}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 225 V dc			90	µA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 300 V dc			90	µA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 375 V dc			90	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low-temperature operation		T_A -55°C.				
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1).	h_{FE2}			
2N6249, 2N6249T1 2N6249T3				5		
2N6250, 2N6250T1 2N6250T3				4		
2N6251, 2N6251T1 2N6251T3				3		
Subgroup 4						
Pulsed response:	3251	Test conditions A except test circuit and pulse requirements in accordance with figure 11.	t_{on}			
Turn-on time		V_{CC} = 200 V dc; I_C = 10 A dc.				
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			2.0	µs
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			2.0	µs
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			2.0	µs
Turn-off time		V_{CC} = 200 V dc; I_C = 10 A dc.	t_{off}			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			4.5	µs
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			4.5	µs
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			4.5	µs
Magnitude of small signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1.0 A dc; f = 1 MHz.	$\|h_{FE}\|$	2.5	15	
Open capacitance open circuit	3236	V_{CE} = 10 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz.	C_{obo}		500	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C; t = 1 s; 1 cycle (see figure 12).				
Test 1		I_C = 10 A dc; V_{CE} = 17.5 V dc				
Test 2		I_C = 5.8 A dc; V_{CE} = 30 V dc				
Test 3		I_C = 0.3 A dc; V_{CE} = 100 V dc				
Test 4		2N6249, 2N6249T1, and 2N6249T3 only I_C = 0.13 A dc; V_{CE} = 200 V dc				
Test 5		2N6250, 2N6250T1, and 2N6250T3 only I_C = 0.09 A dc; V_{CE} = 275 V dc				
Test 6		2N6251, 2N6251T1, and 2N6251T3 only I_C = 0.065 A dc; V_{CE} = 350 V dc				
End-point electrical measurements		Table I, subgroup 2				
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) (see figure 13.) T_C = +25°C; duty cycle ≤ 10 percent; R_S = 0.1.				
Test 1		T_P ≈ 5 ms (vary to obtain I_C); R_{BB1} = 2Ω V_{BB1} = 10 V dc R_{BB2} = 50Ω V_{BB2} = 4 V dc V_{CC} = 10 V dc I_C = 10 A dc L = 50 µH at 10 A dc				
Test 2		T_P ≈ 5 ms (vary to obtain I_C); R_{BB1} = 40Ω V_{BB1} = 10 V dc R_{BB2} = 50Ω V_{BB2} = 4 V dc V_{CC} = 10 V dc I_C = 2.0 A dc L = 500 µH at 2.0 A dc R_L 0.01Ω				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)		(See figure 14) clamped inductive sweep T_C = +25C; duty cycle ≤ 10 percent; T_P ≈ 5 ms (vary to obtain I_C); R_S = 0.1Ω V_{CC} = 10 V dc I_C = 10 A dc Clamp voltage: 2N6249, 2N6249T1 2N6249T3 = 200 V dc 2N6250, 2N6250T1 2N6250T3 = 275 V dc 2N6251, 2N6251T1 2N6251T3 = 350 V dc				
End-point electrical measurements		table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{ces} = 0 V				
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc, L = 42 mH, f = 30 - 60 Hz (see figure 10)	$V_{(BR)CEO}$			
2N6249, 2N6249T1 2N6249T3				200		V dc
2N6250, 2N6250T1 2N6250T3				275		V dc
2N6251, 2N6251T1 2N6251T3				350		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc, L = 14 mH, f = 30 - 60 Hz, R_{BE} = 50 ohms (see figure 10)	$V_{(BR)CER}$			
2N6249, 2N6249T1 2N6249T3				225		V dc
2N6250, 2N6250T1 2N6250T3				300		V dc
2N6251, 2N6251T1 2N6251T3				375		V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		200	µA dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 150 V dc			2.0	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 225 V dc			2.0	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 300 V dc			2.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 225 V dc			20	µA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 300 V dc			20	µA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 375 V dc			20	µA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/- Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 300 V dc			1.0	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 375 V dc			1.0	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 450 V dc			1.0	mA dc
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	[h_{FE1}] 5/			
2N6249, 2N6249T1 2N6249T3				[5]	50	
2N6250, 2N6250T1 2N6250T3				[4]	50	
2N6251, 2N6251T1 2N6251T3				[3]	50	
Collector to emitter saturated voltage	3071	I_C = 10 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			1.725	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			1.725	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			1.725	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			2.59	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			2.59	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			2.59	V dc
Subgroup 2						
Total Dose Irradiation	1019	Gamma exposure				
2N6249, 2N6249T1 2N6249T3		Vces = 160 V				
2N6250, 2N6250T1 2N6250T3		Vces = 220 V				
2N6251, 2N6251T1 2N6251T3		Vces = 280 V				

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc, L = 42 mH, f = 30 - 60 Hz (see figure 10)	$V_{(BR)CEO}$			
2N6249, 2N6249T1 2N6249T3				200		V dc
2N6250, 2N6250T1 2N6250T3				275		V dc
2N6251, 2N6251T1 2N6251T3				350		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc, L = 14 mH, f = 30 - 60 Hz, R_{BE} = 50 ohms (see figure 10)	$V_{(BR)CER}$			
2N6249, 2N6249T1 2N6249T3				225		V dc
2N6250, 2N6250T1 2N6250T3				300		V dc
2N6251, 2N6251T1 2N6251T3				375		V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		200	µA dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 150 V dc			2.0	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 225 V dc			2.0	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 300 V dc			2.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 225 V dc			20	µA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 300 V dc			20	µA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 375 V dc			20	µA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N6249, 2N6249T1 2N6249T3		V_{CE} = 300 V dc			1.0	mA dc
2N6250, 2N6250T1 2N6250T3		V_{CE} = 375 V dc			1.0	mA dc
2N6251, 2N6251T1 2N6251T3		V_{CE} = 450 V dc			1.0	mA dc
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	$[h_{FE1}]$ 5/			
2N6249, 2N6249T1 2N6249T3				[5]	50	
2N6250, 2N6250T1 2N6250T3				[4]	50	
2N6251, 2N6251T1 2N6251T3				[3]	50	
Collector to emitter saturated voltage	3071	I_C = 10 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			1.725	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			1.725	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			1.725	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)}$			
2N6249, 2N6249T1 2N6249T3		I_B = 1.0 A dc			2.59	V dc
2N6250, 2N6250T1 2N6250T3		I_B = 1.25 A dc			2.59	V dc
2N6251, 2N6251T1 2N6251T3		I_B = 1.67 A dc			2.59	V dc

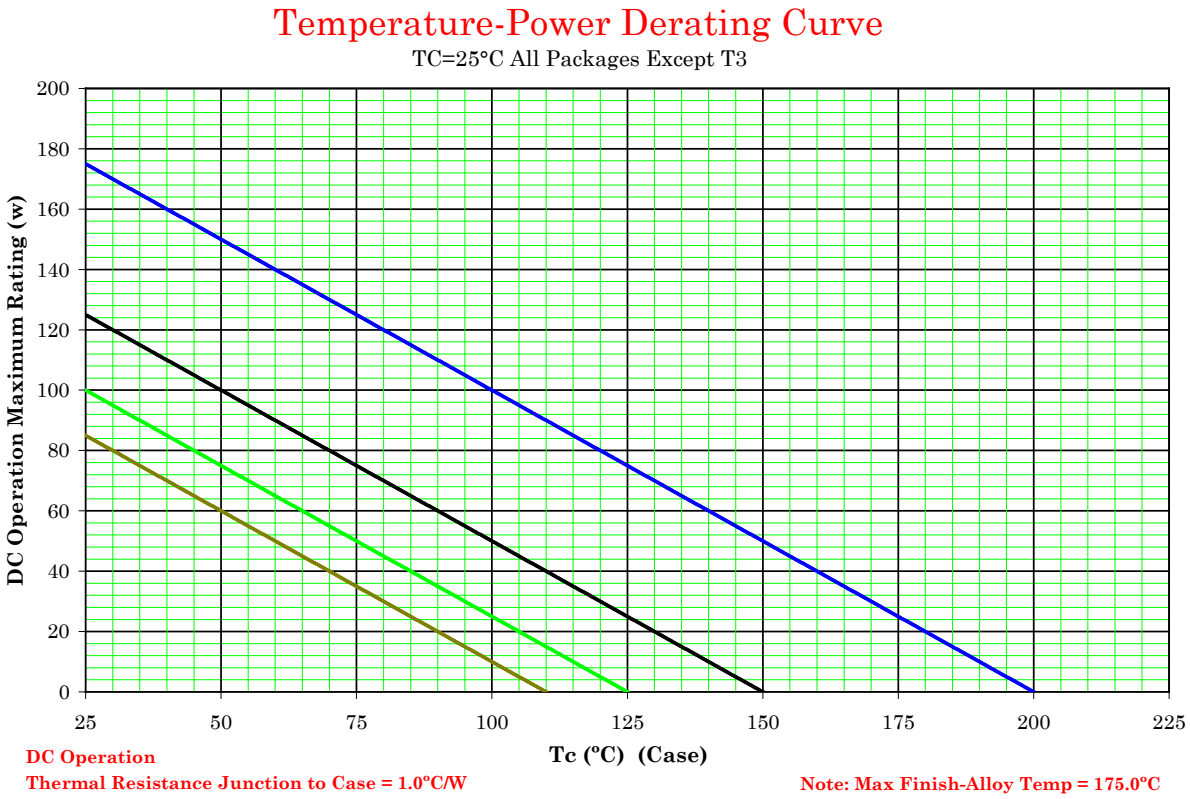
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019, of MIL-STD-750, for how to determine $[h_{FE}]$ by first calculating the delta($1/h_{FE}$) from the pre and post-radiation h_{FE}. Notice that $[h_{FE}]$ is not the same as h_{FE} and cannot be measured directly. The $[h_{FE}]$ value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
End-point electrical measurements		See table I, subgroup 2 .	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles, adjust power or current to achieve ΔT_J = +100°C.	
End-point electrical measurements		See table I, subgroup 2 .	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			15 devices c = 0
Barometric pressure	1001	Condition C, see 1.3, all devices.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, condition B for devices < 400 V.	

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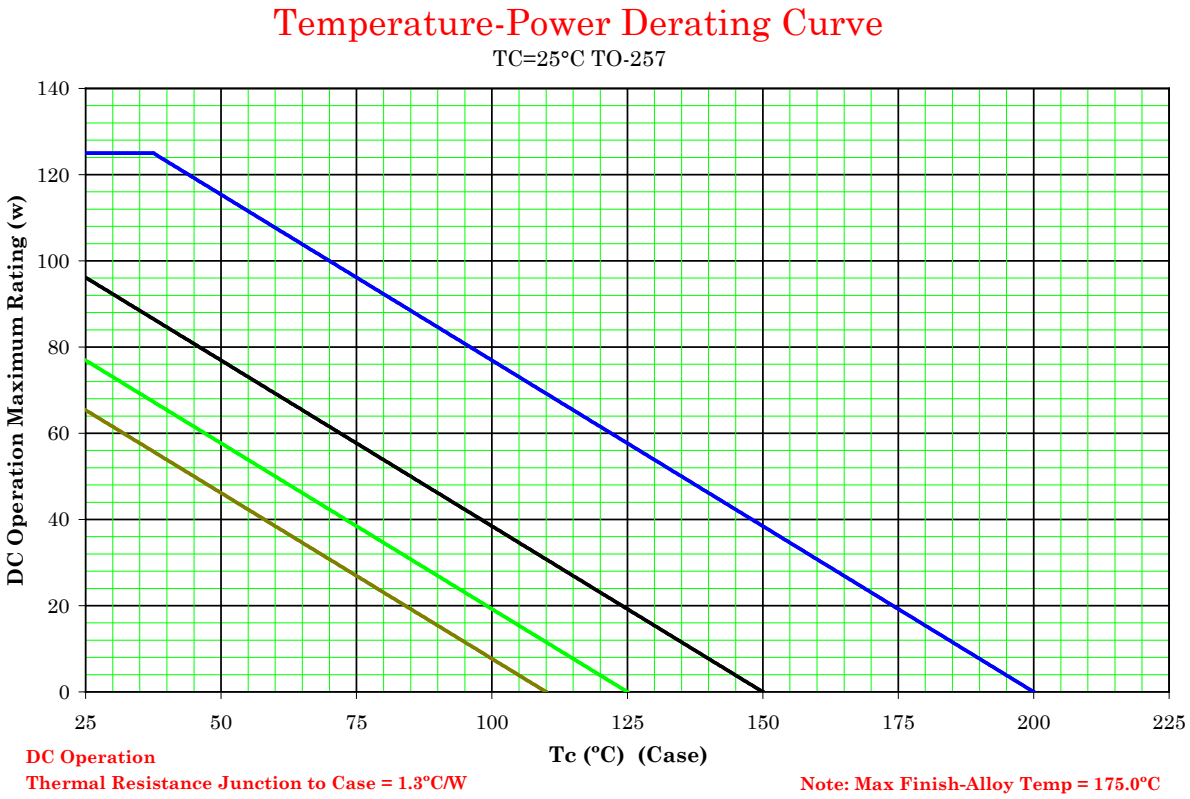


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 5. Derating for all devices except the T3 (TO-257) devices.

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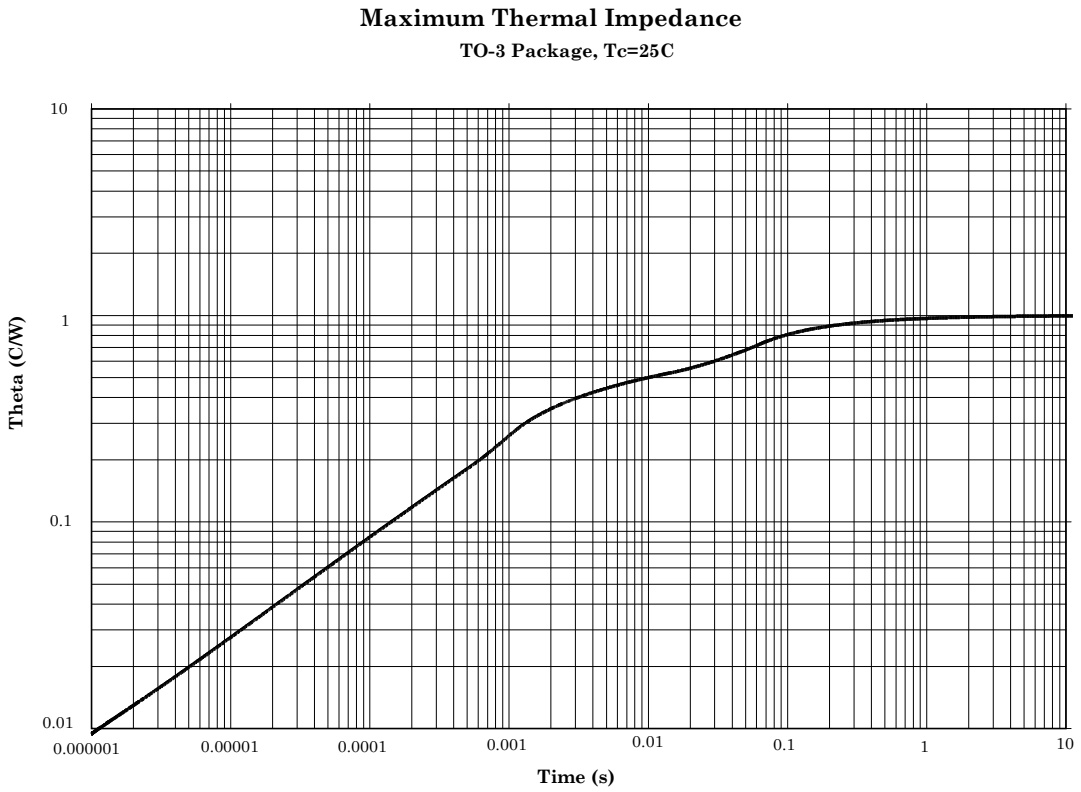


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Derating for T3 (TO-257) packages.

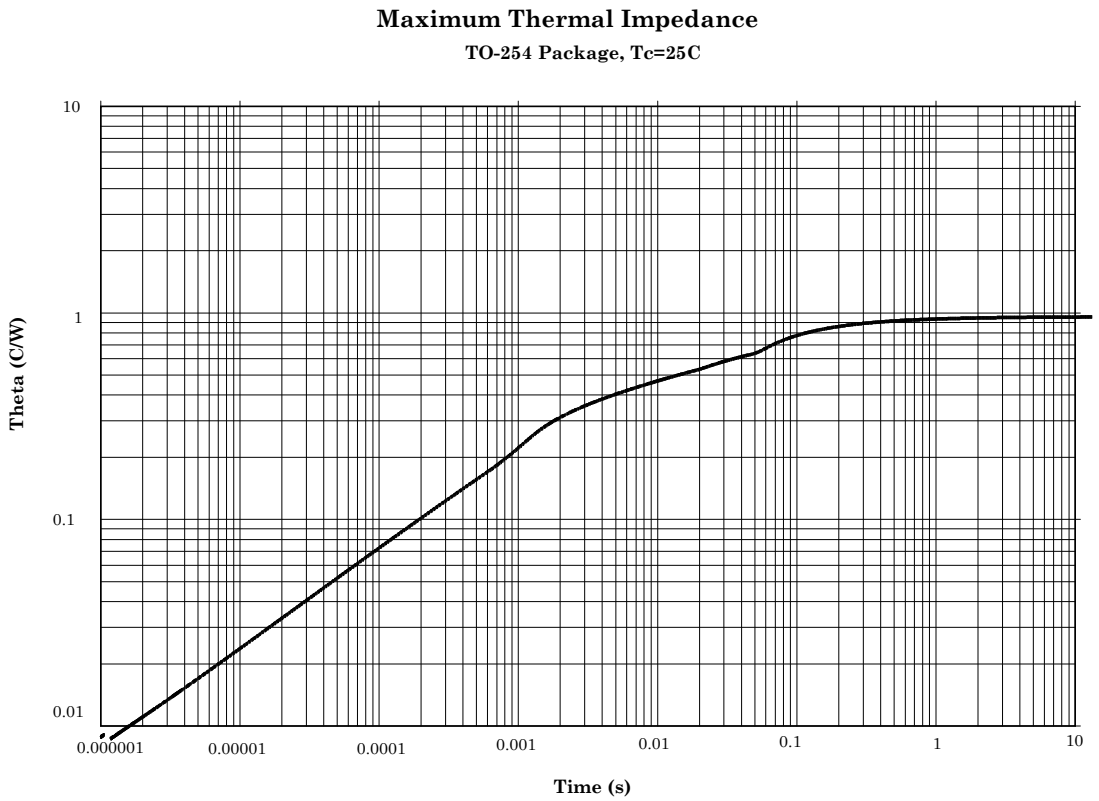
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$R_{\theta JC}$ = 1.0°C/W

FIGURE 7. Thermal impedance graph for 2N6249, 2N6250, and 2N6251 (TO-3).

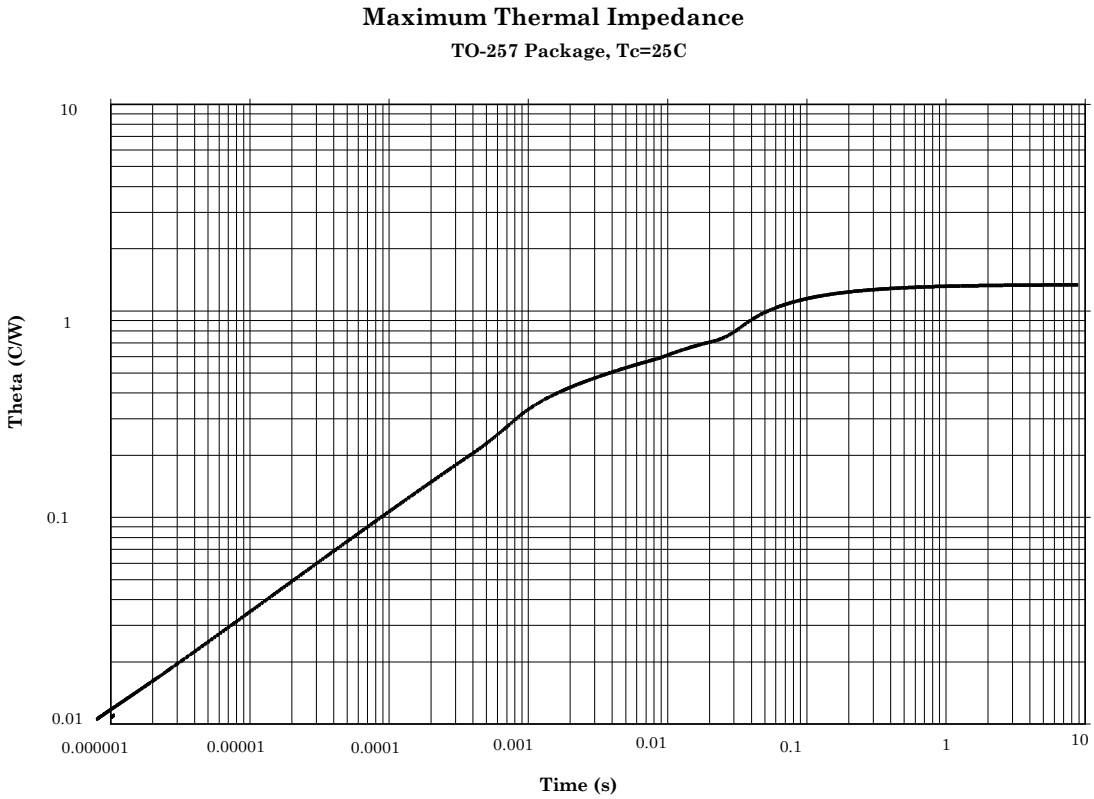
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$R_{\theta JC}$ = 1.0°C/W

FIGURE 8. Thermal impedance graph for 2N6249T1, 2N6250T1, and 2N6251T1 (TO-254AA).

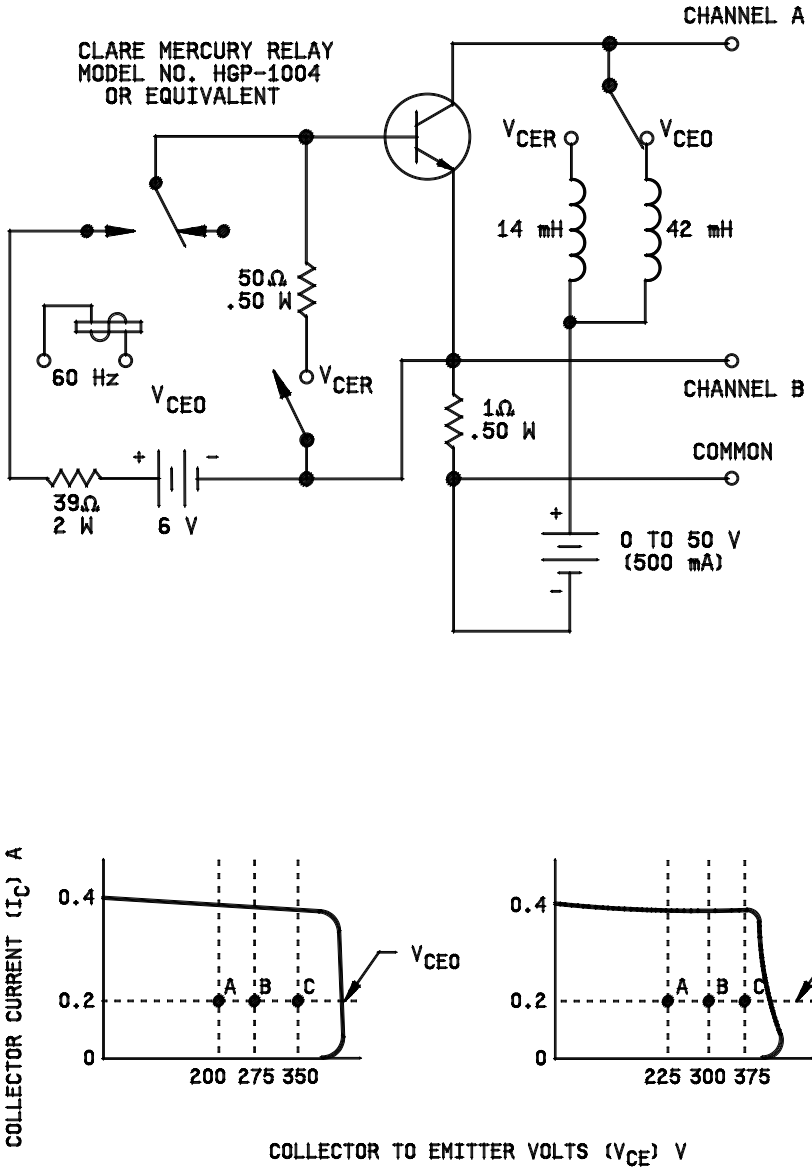
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$R_{\theta JC}$ = 1.3°C/W

FIGURE 9. Thermal impedance graph for 2N6249T3, 2N6250T3, and 2N6251T3 (TO-257AA).

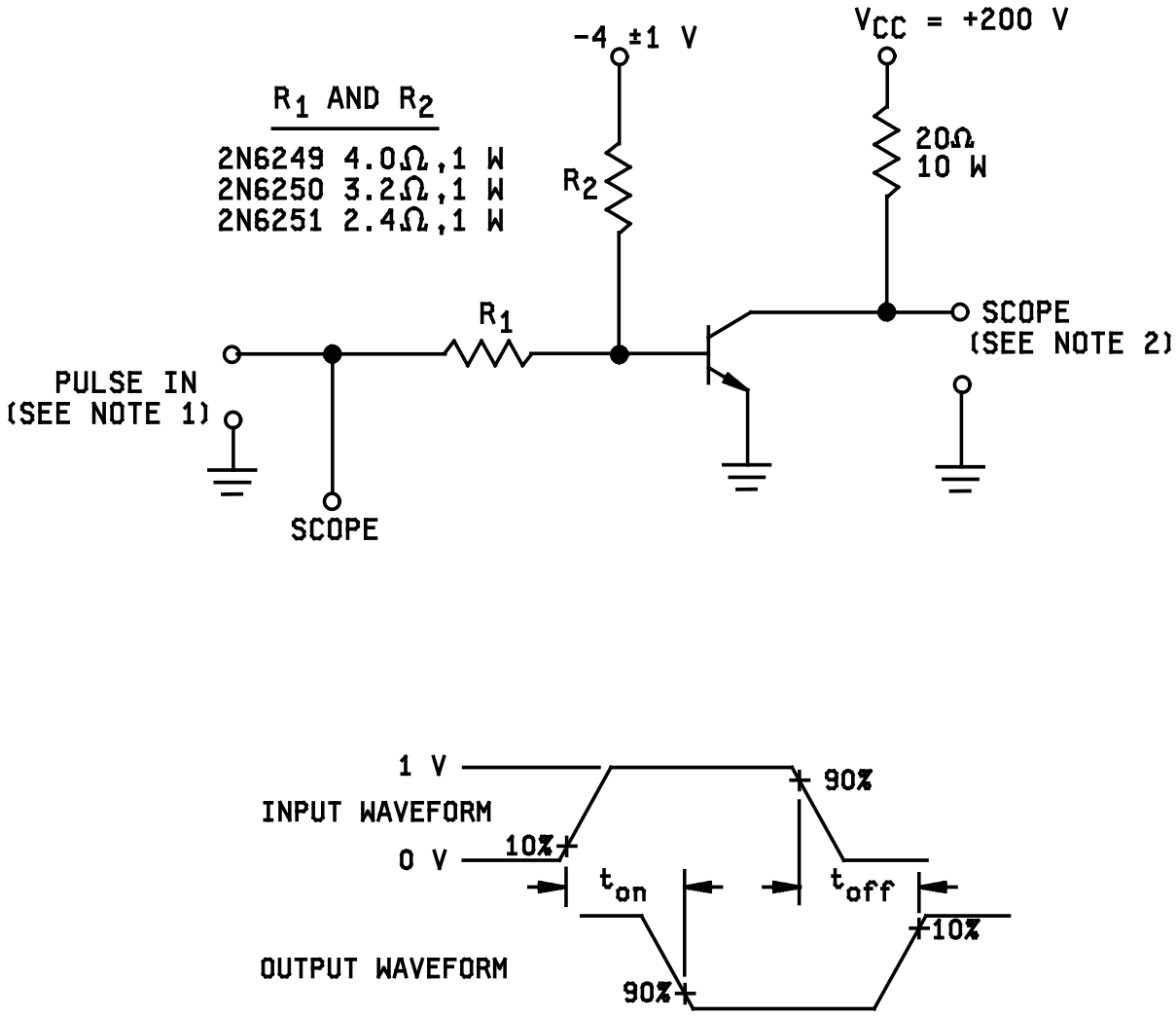
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NOTE: The sustaining voltages V_{CEO} and V_{CER} are acceptable when the traces fall to the right of point “A” for type 2N6249 and 2N6249T1, point “B” for type 2N6250 and 2N6250T1, and point “C” for type 2N6251 and 2N6251T1 (I_C = 0.2 A).

FIGURE 10. V_{CEO}, V_{CER} measurement circuit.

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NOTES:

1. The rise time (t_r) and the fall time (t_f) of the applied pulse shall be each $<$ 20 ns, duty cycle $\leq$ 2 percent; generator source impedance shall be 50 Ω; pulse width = 20 μs.
2. Output sampling oscilloscope; $Z_{in} \geq$ 100 Ωk; $c_{in} \leq$ 50 pF, rise time $<$ 2 ns.

FIGURE 11. Pulse response test circuit.

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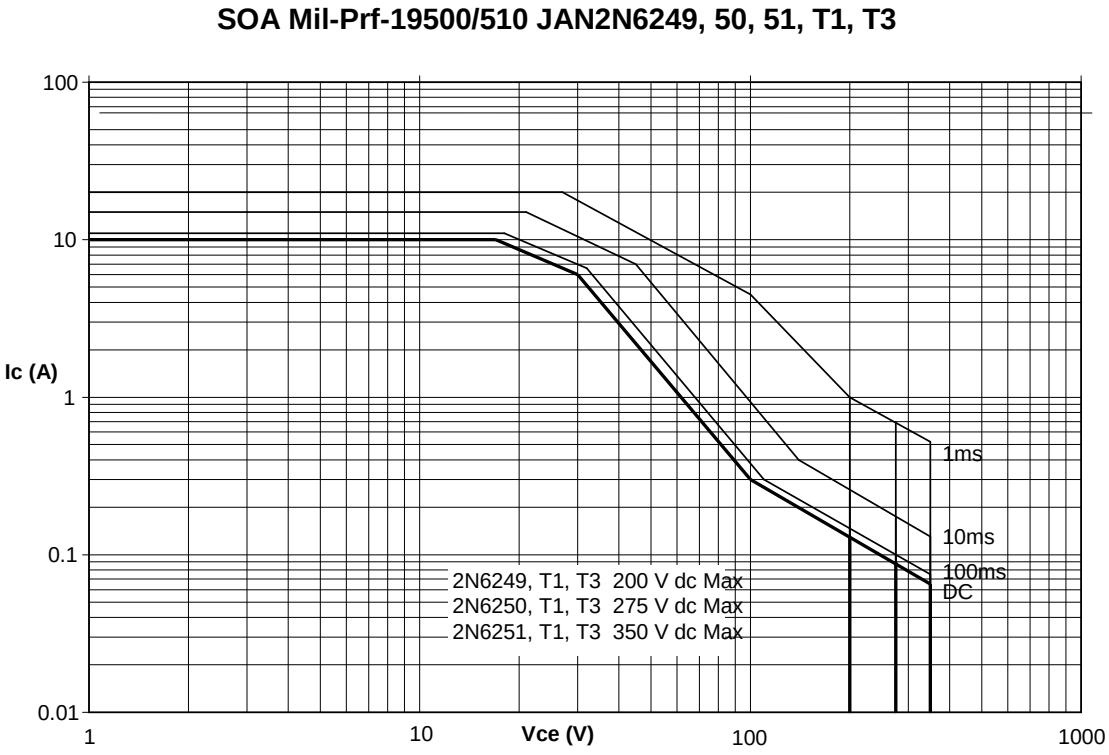


FIGURE 12. Maximum safe operating area graph (continuous dc).

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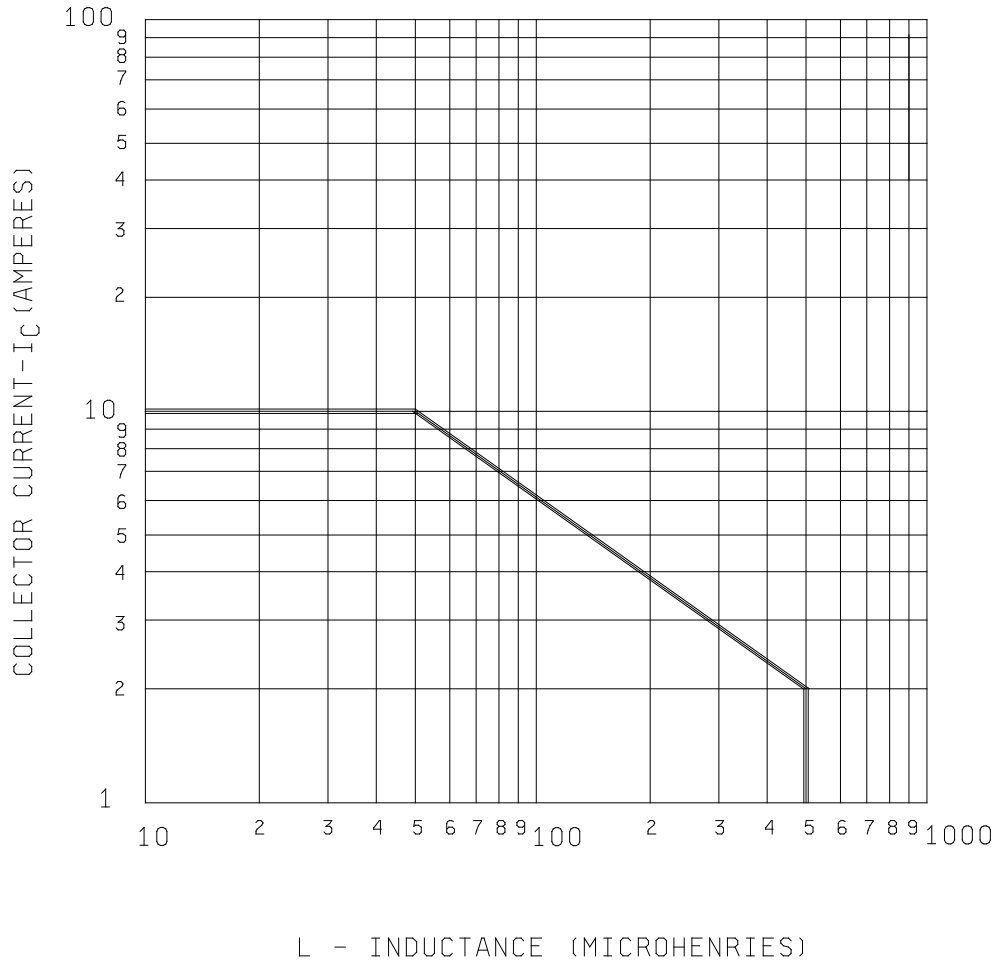
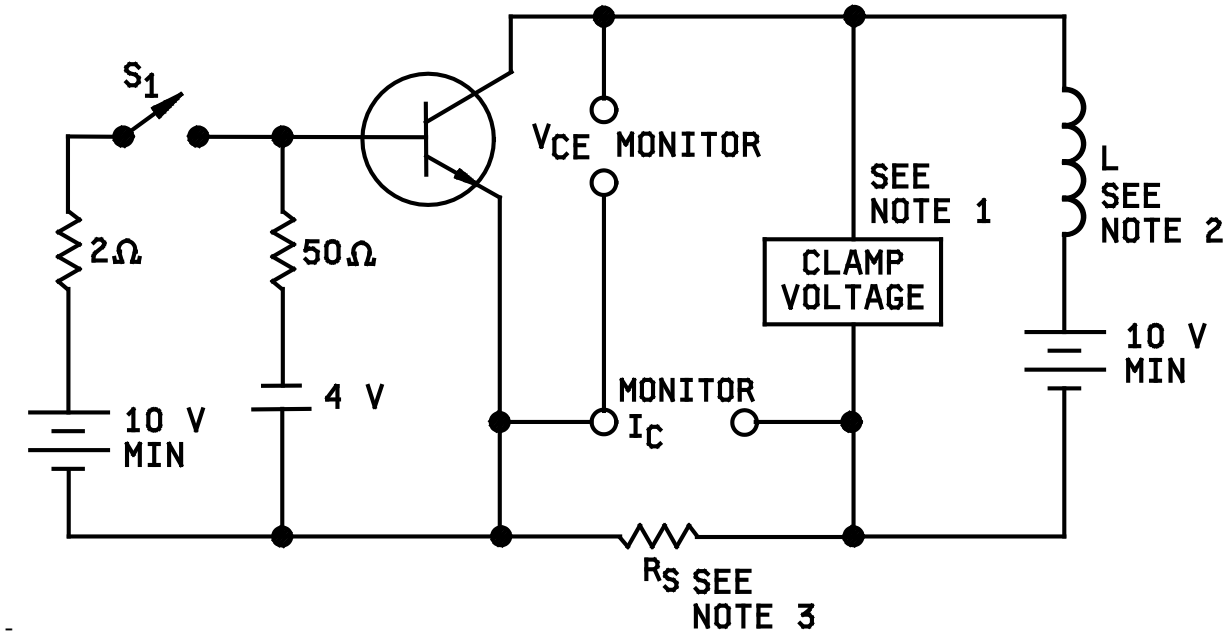


FIGURE 13. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:

1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 50 μH at 10 A with a maximum dc resistance of 0.1 Ω (see 4.5.3).
3. $R_S \leq 0.1\ \Omega$, 12 W, 1 percent tolerance maximum (non-inductive).

Procedure:

1. With switch S_1 closed set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified end-point tests.

FIGURE 14. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N6249) will be identified on the QML.

JANHC and JANKC ordering information		
	Manufacturer	
PIN	33178 (1)	43611
2N6249	A6249	JANHCB2N6249 JANKCB2N6249
2N6250	A6250	JANHCB2N6250 JANKCB2N6250
2N6251	A6251	JANHCB2N6251 JANHCB2N6251

(1) The “A” version of the JANHC and the JANKC part numbers, 2N6249T1, 2N6249T3, 2N6250T1, 2N6250T3, 2N6251T1, and 2N6251T3, were qualified from 29 November 2001; however, they are obsolete at this time.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2014-031)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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